

## TEZPUR, India

WMO: 424150

Lat: 26.617N

Lon: 92.783E

Elev: 259

StdP: 14.56

Time Zone: 5.50 (IND)

Period: 95-18

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 52.3       | 53.8 | 48.6                          | 51.2 | 62.9 | 50.1 | 54.1 | 61.0 | 5.5                   | 66.5 | 5.1 | 66.9 | 0.1                   | N/A  | 0.259 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 9.7                    | 94.4            | 81.7 | 93.0 | 81.3 | 91.6 | 81.0 | 84.2                | 91.0 | 83.5 | 89.9 | 82.8 | 88.9 | 1.0                  | N/A  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 82.6                            | 171.4 | 88.4 | 81.8 | 167.2 | 87.6 | 81.2 | 163.5 | 86.8 | 48.8          | 91.3 | 48.0 | 90.4 | 47.2 | 89.4 | 86.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 5.6               | 4.9  | 3.8 | DB                         | 49.0 | 98.4               | 1.8 | 1.8                                                | 47.7 | 99.7       | 46.7 | 100.7      | 45.6 | 101.7      | 44.3 | 102.9 |
|                   |      |     | WB                         | 48.4 | 85.4               | 1.8 | 0.6                                                | 47.2 | 85.8       | 46.1 | 86.2       | 45.1 | 86.5       | 43.8 | 86.9  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 76.5        | 64.0  | 68.7  | 74.5  | 76.9  | 80.5  | 82.7  | 83.9  | 84.2  | 83.2  | 79.7  | 72.6  | 66.4  | (6)  |
|                                                                        | DBStd       | 7.54        | 2.67  | 3.46  | 3.51  | 3.83  | 3.64  | 3.17  | 2.67  | 2.80  | 2.82  | 3.25  | 2.87  | 2.69  | (7)  |
|                                                                        | HDD50       | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  |
|                                                                        | HDD65       | 74          | 50    | 8     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 16    | (9)  |
|                                                                        | CDD50       | 9666        | 435   | 524   | 759   | 806   | 947   | 981   | 1052  | 1060  | 997   | 920   | 677   | 508   | (10) |
|                                                                        | CDD65       | 4265        | 20    | 112   | 294   | 356   | 482   | 531   | 587   | 595   | 547   | 455   | 227   | 59    | (11) |
|                                                                        | CDH74       | 42533       | 136   | 633   | 2164  | 2684  | 4747  | 6005  | 7130  | 7284  | 6199  | 4040  | 1237  | 274   | (12) |
|                                                                        | CDH80       | 15613       | 3     | 101   | 752   | 943   | 1755  | 2227  | 2876  | 3012  | 2311  | 1371  | 248   | 14    | (13) |
| (14) Wind                                                              | WSAvg       | 0.5         | 0.4   | 0.7   | 1.1   | 1.2   | 0.6   | 0.4   | 0.4   | 0.3   | 0.4   | 0.4   | 0.4   | 0.4   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 76.7        | 0.5   | 1.1   | 2.4   | 6.3   | 10.6  | 14.1  | 13.1  | 12.6  | 10.5  | 4.6   | 1.1   | 0.3   | (15) |
|                                                                        | PrecMax     | 101.1       | 2.8   | 3.8   | 5.9   | 13.0  | 15.6  | 26.5  | 20.0  | 22.0  | 19.6  | 12.1  | 6.7   | 3.1   | (16) |
|                                                                        | PrecMin     | 45.1        | 0.0   | 0.2   | 0.4   | 0.1   | 5.6   | 4.4   | 6.9   | 4.4   | 4.8   | 0.8   | 0.2   | 0.0   | (17) |
|                                                                        | PrecStd     | 16.2        | 0.6   | 0.9   | 1.5   | 2.9   | 2.9   | 5.6   | 3.5   | 4.4   | 3.6   | 2.8   | 1.5   | 0.7   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 80.0  | 86.2  | 92.2  | 95.4  | 95.1  | 95.3  | 94.9  | 95.9  | 95.1  | 93.0  | 86.8  | 81.7  | (19) |
|                                                                        |             | MCWB        | 64.5  | 68.0  | 69.4  | 73.3  | 81.0  | 82.1  | 83.1  | 83.5  | 82.8  | 82.0  | 73.1  | 68.3  | (20) |
|                                                                        | 2%          | DB          | 77.3  | 83.0  | 89.3  | 91.2  | 92.7  | 92.7  | 93.0  | 93.5  | 92.8  | 90.6  | 84.7  | 78.9  | (21) |
|                                                                        |             | MCWB        | 64.1  | 66.4  | 69.7  | 74.0  | 79.6  | 82.2  | 82.7  | 83.5  | 82.3  | 80.1  | 72.4  | 66.5  | (22) |
|                                                                        | 5%          | DB          | 75.1  | 80.6  | 87.0  | 88.2  | 90.5  | 90.8  | 91.3  | 91.8  | 91.0  | 88.6  | 83.0  | 76.9  | (23) |
|                                                                        |             | MCWB        | 63.5  | 65.6  | 69.1  | 73.6  | 78.8  | 81.7  | 82.5  | 83.0  | 81.9  | 78.7  | 71.4  | 65.4  | (24) |
|                                                                        | 10%         | DB          | 72.7  | 78.1  | 84.5  | 85.4  | 88.3  | 88.9  | 89.6  | 89.8  | 89.2  | 86.7  | 81.0  | 75.0  | (25) |
|                                                                        |             | MCWB        | 62.5  | 64.9  | 68.6  | 72.9  | 78.3  | 81.3  | 82.2  | 82.4  | 81.5  | 77.7  | 70.3  | 64.7  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 68.8  | 70.8  | 75.0  | 78.2  | 82.7  | 83.7  | 85.0  | 85.3  | 84.5  | 83.5  | 75.4  | 69.6  | (27) |
|                                                                        |             | MCDB        | 75.4  | 80.4  | 85.5  | 86.7  | 91.8  | 90.9  | 91.8  | 92.3  | 91.9  | 90.4  | 83.5  | 78.4  | (28) |
|                                                                        | 2%          | WB          | 65.7  | 68.6  | 72.8  | 76.5  | 81.2  | 83.0  | 83.7  | 84.4  | 83.4  | 81.7  | 73.7  | 68.0  | (29) |
|                                                                        |             | MCDB        | 73.4  | 78.8  | 83.2  | 85.3  | 89.6  | 89.8  | 90.1  | 90.9  | 90.3  | 88.2  | 82.0  | 76.7  | (30) |
|                                                                        | 5%          | WB          | 64.2  | 67.2  | 71.1  | 75.4  | 80.0  | 82.4  | 83.1  | 83.6  | 82.6  | 80.3  | 72.6  | 66.7  | (31) |
|                                                                        |             | MCDB        | 72.6  | 77.2  | 81.9  | 84.3  | 87.7  | 88.9  | 89.2  | 89.6  | 88.9  | 86.1  | 80.5  | 74.5  | (32) |
|                                                                        | 10%         | WB          | 62.9  | 65.9  | 69.9  | 74.2  | 79.1  | 81.7  | 82.4  | 82.8  | 81.8  | 79.0  | 71.5  | 65.4  | (33) |
|                                                                        |             | MCDB        | 71.0  | 75.5  | 80.9  | 82.4  | 86.2  | 87.5  | 88.3  | 88.3  | 87.5  | 84.6  | 78.8  | 72.8  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 19.7  | 19.9  | 19.5  | 14.8  | 12.9  | 9.9   | 9.4   | 9.7   | 10.7  | 14.0  | 17.6  | 19.3  | (35) |
|                                                                        |             | MCDBR       | 23.1  | 23.6  | 24.3  | 20.8  | 16.8  | 13.1  | 12.4  | 12.7  | 14.0  | 16.1  | 19.7  | 21.2  | (36) |
|                                                                        | 5% WB       | MCWBR       | 11.1  | 9.1   | 8.8   | 7.4   | 6.5   | 5.1   | 4.8   | 5.0   | 5.6   | 6.9   | 8.6   | 9.9   | (37) |
|                                                                        |             | MCWBR       | 21.1  | 20.5  | 20.6  | 16.3  | 14.8  | 11.5  | 11.3  | 11.8  | 12.5  | 14.0  | 17.3  | 19.0  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.552 | 0.580 | 0.633 | 0.644 | 0.587 | 0.539 | 0.467 | 0.474 | 0.481 | 0.454 | 0.484 | 0.500 | (40) |
|                                                                        |             | taud        | 1.812 | 1.778 | 1.701 | 1.747 | 1.921 | 2.095 | 2.324 | 2.297 | 2.254 | 2.280 | 2.076 | 1.996 | (41) |
|                                                                        | Ebn at Noon |             | 214   | 221   | 219   | 222   | 235   | 245   | 263   | 261   | 256   | 256   | 235   | 224   | (42) |
|                                                                        |             | Edh at Noon | 60    | 66    | 75    | 74    | 62    | 52    | 41    | 42    | 43    | 40    | 45    | 48    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1136  | 1331  | 1498  | 1505  | 1530  | 1389  | 1416  | 1420  | 1350  | 1353  | 1271  | 1133  | (44) |
| (45)                                                                   | RadStd      |             | 82    | 99    | 115   | 139   | 121   | 109   | 123   | 134   | 91    | 101   | 80    | 58    | (45) |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (2 neighbors) | +0.99 | N/A     | +1.49  | +1.44   | +1.39 | +1.40 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page